

## Task Questions

### 1. Question 1

**Answer:** I would place it in the private subnet for enhanced security so only resources from the Virtual Private Cloud can access it and prevents any unauthorized access from the external network (internet). This feature is only beneficial if my RDS instance doesn't need much exposure from the external network to minimize any potential threats.

### 2. Question 2

**Answer:** In the AWS Management Console, I will navigate to the Security Group associated with the RDS instance.

In that we need to create a new inbound rule for the security group, like assigning port ranges that should be accessed, and specify the source IP address or range that our application server is hosted on.

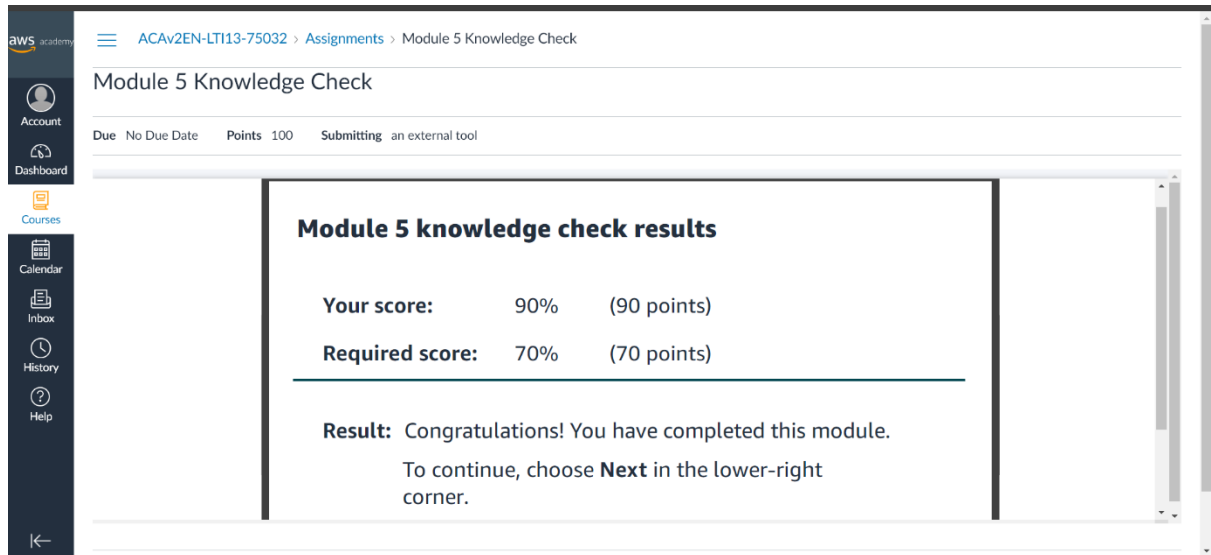
By specifying these rules, we can make sure that the security group will only allow inbound traffic on the specified port only from the designated port only from the designated source, ensuring that only your application servers can connect to the RDS instance.

## Quiz

### 1. Knowledge Check for Module 5

This knowledge check was comprised of the most important questions throughout the module 5. The questions asked was kind of a summary to the whole module 5. In this knowledge check I was able to see my understanding of this module, and the from the mistakes I was able to correct myself of the parts that I was not sure about.

This knowledge check was mainly about the relational and non-relational databases. There were questions about best suited database models according to the company needs, the types of databases, the benefits of each type, the use cases that each type of database is good at, and techniques used by the them and how they are used to migrate the on-premise data into cloud.



The screenshot shows the AWS Academy interface for the 'Module 5 Knowledge Check'. The left sidebar contains navigation links: Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area displays the following information:

- Module 5 Knowledge Check**
- Due:** No Due Date
- Points:** 100
- Submitting:** an external tool

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**Module 5 knowledge check results**

|                        |     |             |
|------------------------|-----|-------------|
| <b>Your score:</b>     | 90% | (90 points) |
| <b>Required score:</b> | 70% | (70 points) |

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**Result:** Congratulations! You have completed this module.  
To continue, choose **Next** in the lower-right corner.

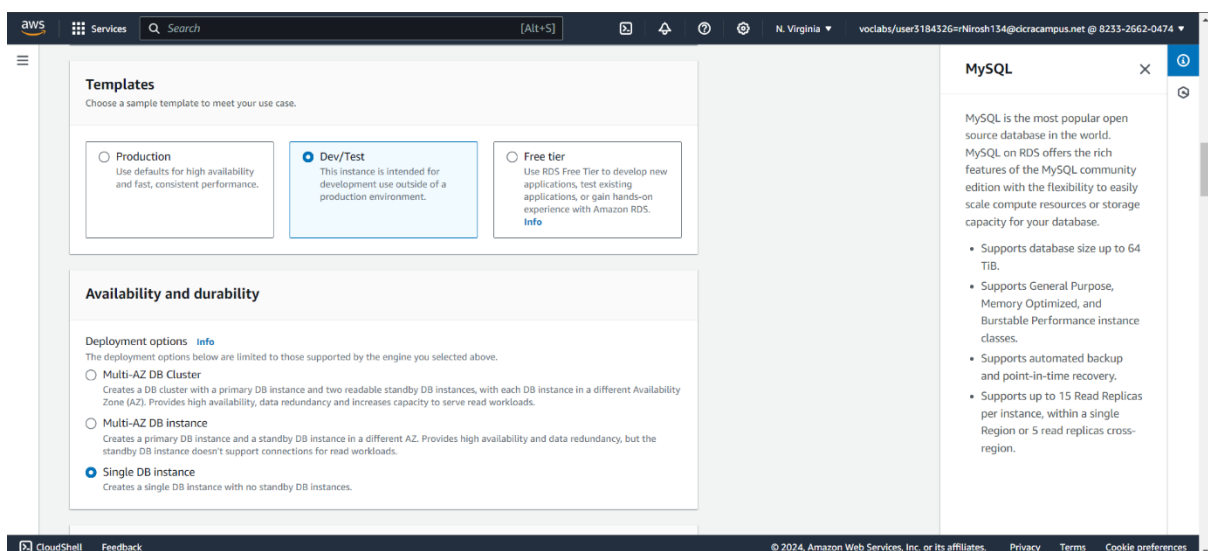
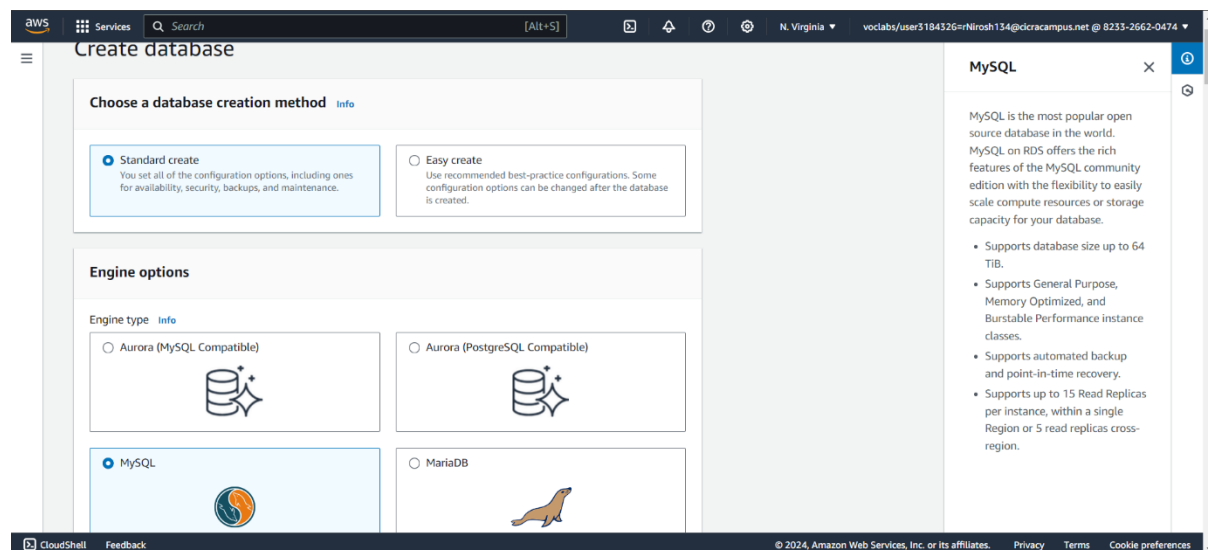
## Module 5: Guided Lab – Creating an Amazon RDS Database

### Task 1: Creating an Amazon RDS database

In the RDS console, I searched for "RDS" in the services search box and selected it. Then, clicked on "Create database."

Under the "Engine options," I chose MySQL from the available options.

I Selected "Dev/Test" under the Templates section. Then Chose "Single DB instance" under the Availability and durability section.



## SIT233 – Cloud Computing

### Task 4.1P: Creating an Amazon RDS database

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Then I configured some settings:

Set the DB instance identifier as "inventory-db."

Username as "admin."

Password as "lab-password" (confirm it).

Under DB instance class, select "Burstable classes (includes t classes)" and then choose "db.t3.micro."

For Connectivity, set the VPC to "Lab VPC" and select the existing VPC security group "DB-SG." Remove the default security group.

Set the initial database name as "inventory."

Clear (turn off) the "Enable Enhanced monitoring" option.

The screenshot shows the 'Settings' page in the AWS Management Console. The 'DB instance identifier' is set to 'inventory-db'. Under 'Credentials Settings', the 'Master username' is 'admin'. The 'Credentials management' section shows 'Self managed' selected. The 'Master password' field is visible with masked characters. A sidebar on the right provides information about MySQL on RDS.

**Settings**

**DB instance identifier** [Info](#)  
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.  
  
The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

**▼ Credentials Settings**

**Master username** [Info](#)  
Type a login ID for the master user of your DB instance.  
  
1 to 16 alphanumeric characters. The first character must be a letter.

**Credentials management**  
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*  
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed  
Create your own password or have RDS create a password that you manage.

☐ Auto generate password  
Amazon RDS can generate a password for you, or you can specify your own password.

**Master password** [Info](#)

**MySQL**

MySQL is the most popular open source database in the world. MySQL on RDS offers the rich features of the MySQL community edition with the flexibility to easily scale compute resources or storage capacity for your database.

- Supports database size up to 64 TiB.
- Supports General Purpose, Memory Optimized, and Burstable Performance instance classes.
- Supports automated backup and point-in-time recovery.
- Supports up to 15 Read Replicas per instance, within a single Region or 5 read replicas cross-region.

The screenshot shows the 'Instance configuration' page. The 'DB instance class' is set to 'db.t3.micro'. The 'Storage' section is partially visible. A sidebar on the right provides information about MySQL on RDS.

**Instance configuration**  
The DB instance configuration options below are limited to those supported by the engine that you selected above.

**DB instance class** [Info](#)

**▼ Hide filters**

☒ Show instance classes that support Amazon RDS Optimized Writes [Info](#)  
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☒ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

2 vCPUs 1 GiB RAM Network: 2,085 Mbps

**Storage**

**Storage type** [Info](#)

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The screenshot shows the 'Connectivity' page. The 'Compute resource' section has 'Don't connect to an EC2 compute resource' selected. The 'Virtual private cloud (VPC)' section is partially visible. A sidebar on the right provides information about MySQL on RDS.

**Connectivity** [Info](#)

**Compute resource**  
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ Don't connect to an EC2 compute resource  
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

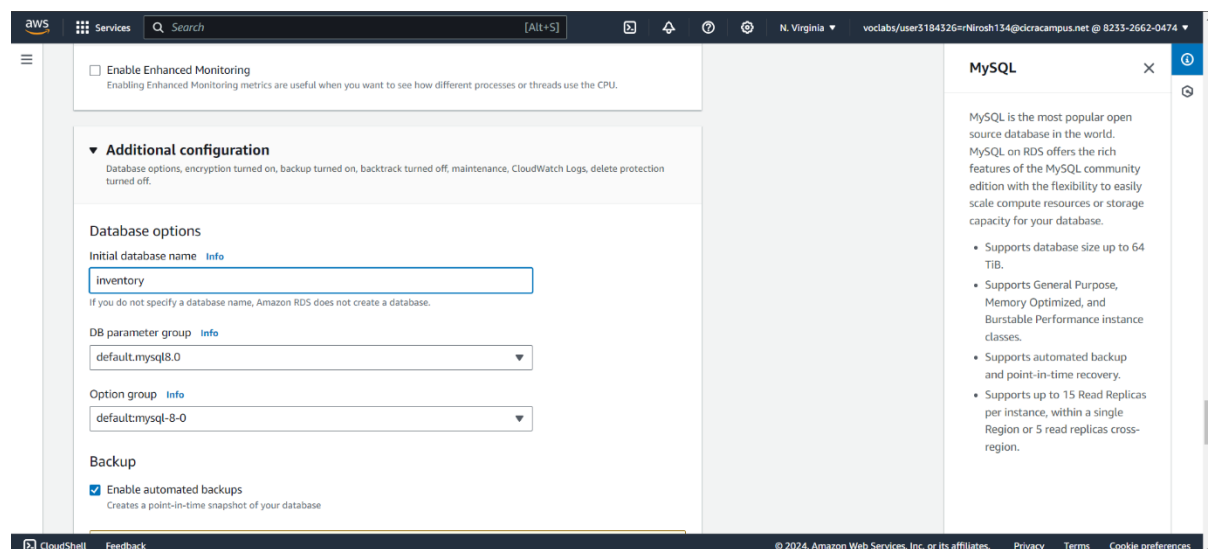
☐ Connect to an EC2 compute resource  
Set up a connection to an EC2 compute resource for this database.

**Virtual private cloud (VPC)** [Info](#)  
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

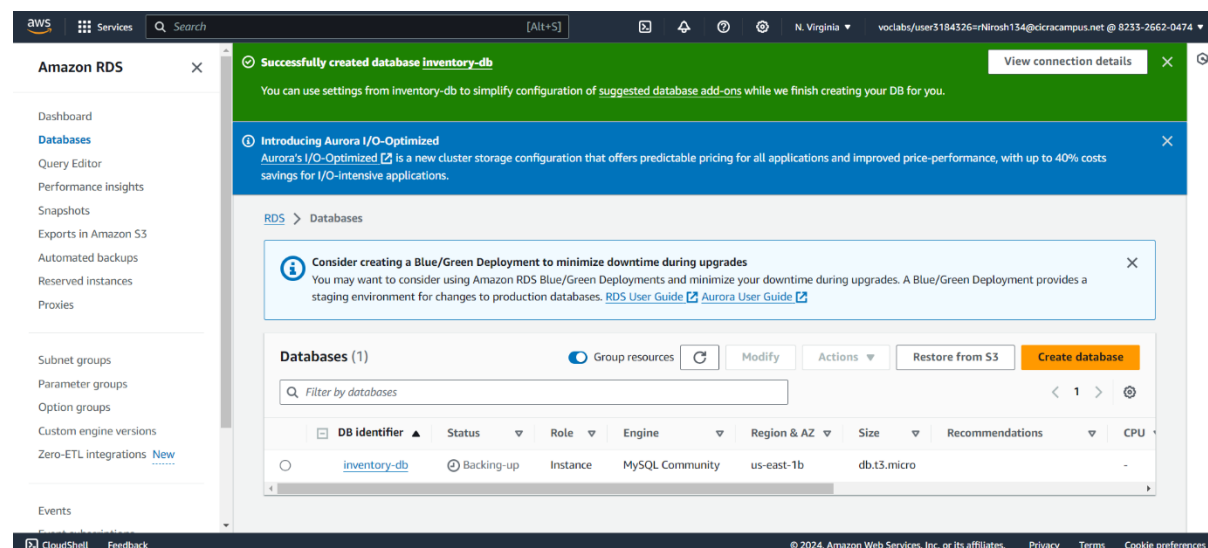
**MySQL**

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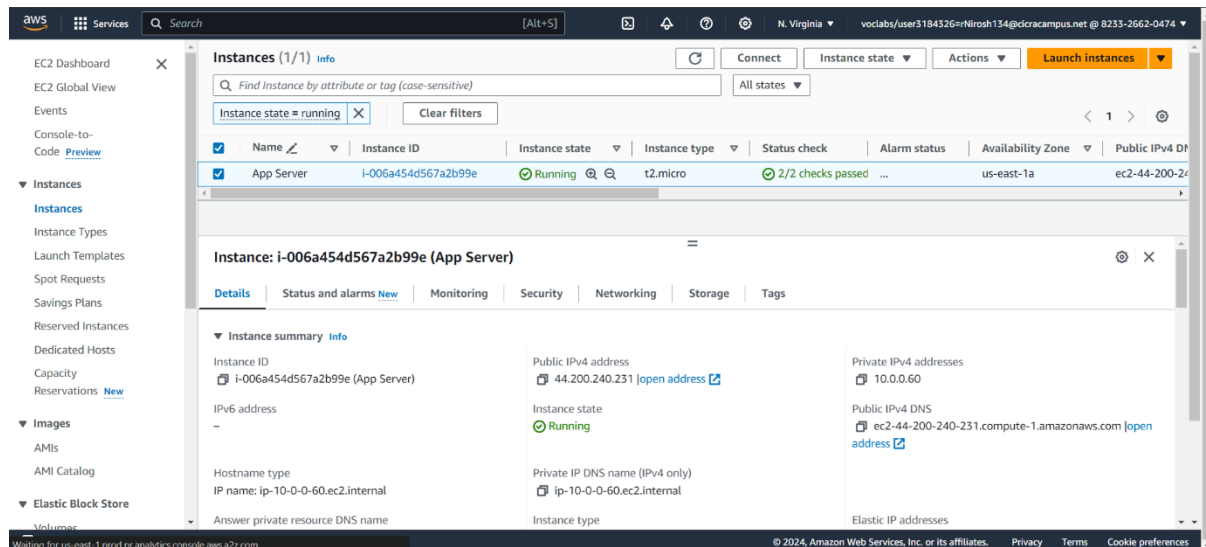


Then at last I clicked the create database and waited until the database was created. Finally, I received a message of successful database creation.

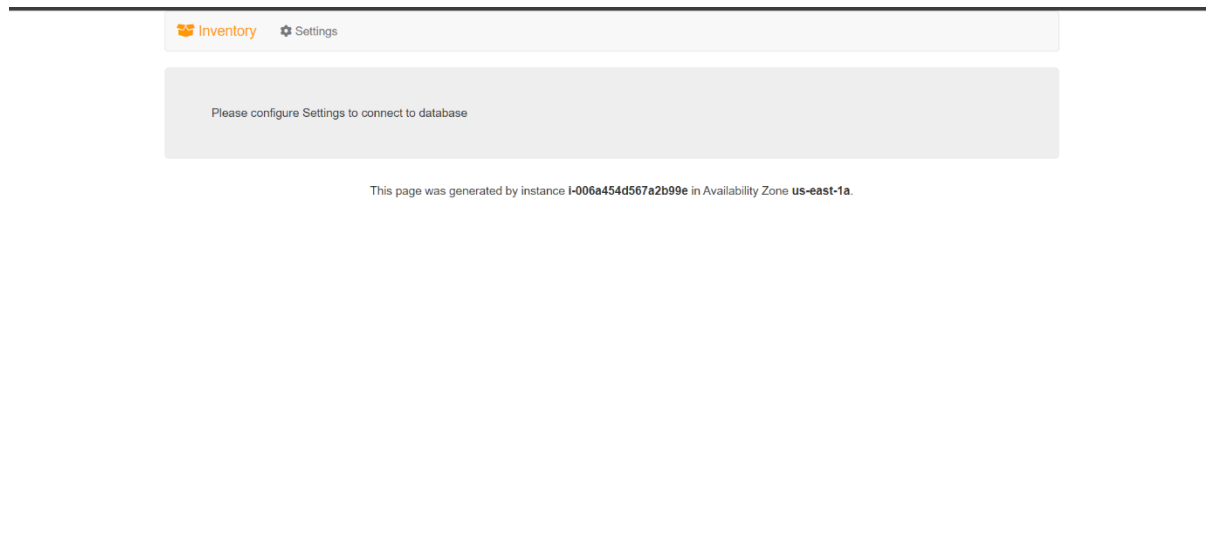


## Task 2: Configuring web application communication with a database instance

I search for "EC2" in the services search box and choose EC2 to open the EC2 console. In the left navigation pane, I select "Instances." In the center pane, I locate a running instance named "App Server." I select the "App Server" instance and navigate to the "Details" tab. Hovering over the Public IPv4 address, I click on the copy icon to copy the address.



Opening a new web browser tab, I paste the copied IP address into the address bar and press ENTER. The web application loads, displaying limited information as it's not yet connected to the database.



Within the web application, I choose the "Settings" option to configure it to use the RDS DB instance created earlier.

Returning to the AWS Management Console without closing the application tab, I search for and choose RDS to open the RDS console. In the left navigation pane, I select "Databases" and choose the "inventory-db." Scrolling to the "Connectivity & Security" section, I copy the Endpoint to my clipboard.

Back in the browser tab with the Inventory application, I enter the following values:

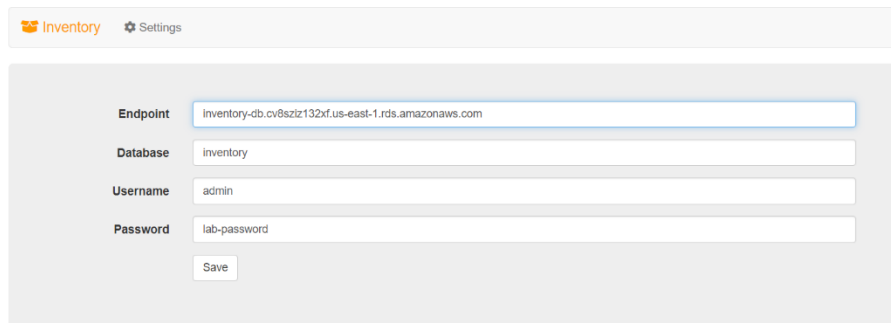
Endpoint: I paste the endpoint I copied earlier.

Database: I enter "inventory."

Username: I input "admin."

Password: I type in "lab-password."

After entering the details, I choose "Save."



The screenshot shows the 'Inventory' application's 'Settings' page. At the top, there is a navigation bar with 'Inventory' and 'Settings' (indicated by a gear icon). Below this, the 'Endpoint' field is populated with 'inventory-db.cv8sziz132xf.us-east-1.rds.amazonaws.com'. The 'Database' field contains 'inventory', the 'Username' field contains 'admin', and the 'Password' field contains 'lab-password'. A 'Save' button is located at the bottom of the form.

With the provided database details, the application now connects to the database, loads initial data, and displays information.

I can now add, edit, and delete inventory information using the web application. Any changes made are stored in the Amazon RDS MySQL database created earlier, ensuring data integrity and availability even if there's a failure in the application server. I insert new records into the table to ensure there are at least 5 or more inventory records before completing the task.

Inventory

Settings

|  | Store       | Item        | Quantity |
|--|-------------|-------------|----------|
|  | Puerto Rico | Amazon Echo | 12       |
|  | Paris       | Amazon Dot  | 3        |
|  | Detroit     | Amazon Tap  | 5        |
|  | Sri Lanka   | Amazon S3   | 12       |
|  | North Korea | Amazon EC2  | 36       |
|  | Mexico      | Amazon RDS  | 25       |

+ Add Inventory

This page was generated by instance `I-006a454d567a2b99e` in Availability Zone `us-east-1a`.

Grade:

My Total score is 20/20

aws academy

Account

Dashboard

Courses

Calendar

Inbox

History

Help

ACAv2EN... > Assignments  
> Module 5 Guided Lab - Creating an Amazon RDS Database

Module 5 Guided Lab - Creating an Amazon RDS Database

Due No Due Date Points 100 Submitting an external tool

Submit Details AWS Start Lab End Lab 2:16 Instructions Grades Actions

Files README Terminal Source

EN\_US

Manager console, under **Parameter Store**.

Submitting your work

28. At the top of these instructions, choose **Submit** to record your progress and when prompted, choose **Yes**.

29. If the results don't display after a couple of minutes, return to the top of these instructions and choose **Grades**.

Tip: You can submit your work multiple times. After you change your

Launch Terminal

Total score 20/20

[Task 1A] inventory-db found 5/5

[Task 1B] Database engine type 5/5

[Task 1C] Database instance type 5/5

[Task 2] Add Inventory Records 5/5